Spatial and Behavioral Variation in Bike-Share Trip Duration: A study of Bluebike system in Greater Boston

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1 Introduction

Background of research

The prevalence of bike usage has recently increased due to the widespread popularity of bike-sharing programs in major metropolitan cities. Issues like traffic congestion and environmental pollution are becoming more pressing in the world because of fast urbanization and expanding automobile ownership. However, bike sharing, being an eco-friendly and low-carbon transportation approach, substantially enhances sustainable mobility by minimizing automobile emissions while facilitating active transportation (Fishman, 2016). Bike-sharing initiatives such as Bluebikes (Boston), Citi bike (NYC), Bay wheels (San Francisco), Metro bike share (Los Angeles) etc. have grown in popularity in recent years. The increased awareness of their environment footprint led to reduce traffic and carbon emission in cities, enhance air quality, improve physical health and promote climate-resilient urban transportation systems (Cohen & Kietzmann, 2014; Jäppinen et al., 2013; Otero et al., 2018).

Motivation of the research

Comprehending BlueBikes trip patterns is crucial for enhancing service quality and optimizing policy and business insights. The duration of trips, station utilization, and rider classification (Member vs. Customer) reveal how individuals interact with the bike-share system and the determinants that shape their travel decisions. These data facilitate the identification of demand peaks, variations in trip lengths across neighborhoods, and the differing motivations of casual vs. frequent riders (Shaheen & Chan, 2016).

This analysis is especially important from a policy and business perspective. By understanding rider behavior, BlueBikes can optimize station placement, adjust fleet rebalancing schedules, and design pricing schemes that reflect actual usage patterns. For example, regions with substantial tourist activity naturally exhibit more Customers and fewer Subscribers (Kaviti et al., 2019a). Policy makers may leverage these insights to support equitable access, ensure coverage in underserved areas, and align bike-share availability with broader transportation goals, while the company can calibrate per-ride fees, subscription offerings, and loyalty incentives to match localized demand (Kaviti et al., 2019b; Shaheen et al., (Kaviti et al., 2019b; Shaheen et al., 2014).

Overall, these insights underscore the necessity of integrating data-driven improvements into station location, price, and service coverage to formulate policies that fairly address the mobility requirements of varied rider demographics.

Literature review

Bikesharing behavior is often influenced by complex, nonlinear relationships among built environment factors, weather conditions, and rider characteristics patterns that linear models cannot fully capture. Using nonlinear analytical methods allows researchers to uncover hidden thresholds and interaction effects that more accurately explain trip frequency, duration, and station demand (Liu et al., 2025).

Evidence indicates how polynomial regression, combined with K-means clustering, can effectively model hourly station-level demand by transforming trip, weather, and temporal features into a second-order polynomial space. This allows the model to learn curvilinear and interaction effects that linear methods overlook. Model comparison shows that second-order polynomial regression offers the optimum mix of complexity and accuracy, surpassing both linear and higher-order polynomial models due to reduced overfitting. These findings demonstrate the effectiveness of polynomial feature expansion in capturing nonlinear demand structures in bikesharing systems (Rylander et al., 2014).

Recent research on rental bike demand prediction demonstrates that regression-based techniques may accurately forecast how environmental and temporal factors influence hourly bike usage. Multiple linear regression and second-degree polynomial regression have been used to estimate hourly rental counts, with polynomial models providing more explanatory power due to their ability to capture curved trends between climate variables and demand. However, despite great fit statistics, polynomial models in this context tend to overfit. Overall, the literature emphasizes the need of including nonlinear factors when predicting rental bike demand, as well as the need for careful evaluation of model complexity to balance accuracy and generalizability (Kumar et al., 2023).

Another study models cab pricing using correlation analysis and a third-degree polynomial regression, which is applied to capture the nonlinear relationship between price and distance, where the curve bends upward as distance increases. This polynomial approach reveals pricing inflection points and slope changes that linear regression cannot detect, showing that fare increases accelerate over longer distances and vary by cab type. The findings highlight that third-degree polynomial regression is essential for modeling real-world cab pricing because it captures curved, nonlinear dynamics that reflect true fare structures and improves predictive reliability (Zihao et al., 2023).

Apart from bike sharing, another study forecasts traffic congestion by integrating various factors average entry/exit waiting time, day of the week, hour of the day, holidays, and precipitation rate and evaluates multiple models, including simple linear regression, polynomial regression, radial basis function neural networks, and feedforward neural networks. The findings indicate that polynomial regression more effectively captures nonlinear correlations between traffic characteristics and congestion compared to linear regression; nonetheless, a feedforward neural network surpasses all methods in predicting accuracy, achieving 97.6%. The article indicates that although polynomial regression enhances linear models by capturing curvature and interaction effects, neural network models may still be superior for optimal accuracy and robustness across diverse traffic situations (Fahs et al., 2023). Since this course did not cover Deep Neural Networks, this study will use a third-degree polynomial regression model.

Research Question

Does rider type (Subscriber and Customer) influence BlueBikes trip duration across five Greater Boston cities, and how does this relationship vary when modeled using degree-3 polynomial regression?

2 Methods

2.1 Dataset

The study uses BlueBikes Trip Data (https://bluebikes.com/system-data) to analyze trip duration, rider types, and temporal usage patterns across the greater Boston area using R-programming. To ensure accurate geographic information, Google Maps Geocoding API (https://developers.google.com/maps/documentation) is used to correct missing or invalid ZIP codes associated with each ride. Cleaned ZIP codes are then matched to their corresponding Massachusetts cities

using the USPS ZIP Code Lookup tool (https://tools.usps.com/zip-code-lookup.htm), allowing appropriate handling of missing values and improving spatial accuracy in the dataset.

2.2 Exploratory data analysis

To conduct this study, 2022 BlueBikes ridership patterns are analyzed. The dataset contains *3,757,281* observations condensed across 14 features, each detailing a specific characteristic of a BlueBikes trip. The features of primary interest from **(Table 1)** are Trip Duration; Start and Stop Times; Start and End Station Longitude and Latitude; User type; and Postal Code. These variables provide both temporal and spatial dimensions that help characterize trip-level behavior.

***Table 1**: Variable name and description from Bluebike dataset*

Variable	Type	Unit	Description
Trip duration	integer	seconds	Total length of bike trips
Start time	character	yyyy-mm-dd hh:mm:ss	Date and time when ride started
Stop time	character	yyyy-mm-dd hh:mm:ss	Date and time when ride ended
Start station id	integer	integer	Unique identifier assigned to the bike station where the trip started
Start station name	character	text	Starting bike station name
Start station latitude	numeric	decimal degrees	Latitude coordinate of the trip's starting station
Start station longitude	numeric	decimal degrees	Longitude coordinate of the trip's starting station
End station id	integer	integer	Unique identifier assigned to the bike station where the trip ended
End station name	character	text	Ending bike station name
End station latitude	numeric	decimal degrees	Latitude coordinate of the trip's ending station
End station longitude	numeric	decimal degrees	Longitude coordinate of the trip's ending station
Bike id	integer	integer	Unique ID assigned to each bike used in the rental system
User type	character	subscriber/customer	Subscriber: long term registered member; Customer: short term casual rider
Postal code	character	5 digit zip code	Postal code of the start or end area

2.2.1 Data cleaning and pre-processing

2.2.1.1 Cleaning Missing ZIP codes and duplicates in Bluebike data

No duplicates were found for the variables of interest discussed above. However, *465,637* rides lacked ZIP codes, and several existing ZIPs (e.g., 19454, 08802, 18064) were incorrect because they fall outside the Greater Boston region. To correct this, the Google Maps Geocoding API was used to assign ZIP codes to all start and end stations based on their latitude and longitude. The BlueBikes reported ZIPs were then compared with the ZIPs generated using Google Geocoding.

To perform this comparison, we created a new variable called "Postal Source". To that end, ZIP codes beginning with **01** or **02** were classified as valid Massachusetts ZIP codes **(Appendix, Table 1)**. Among these, entries that did not match either the rider's start or end station were labeled **Valid but Unmatched**, while all non-Massachusetts ZIP codes were categorized as **Invalid and Unmatched**. Riders whose postal codes aligned with their start station were assigned the postal source **Start Station ZIP**, and those aligning with their end station were assigned **End Station ZIP**.

For observations that were **Missing**, **Valid but Unmatched**, or **Invalid and Unmatched**, the postal code was imputed using the rider's **start-station ZIP**. This choice is supported by the fact that **26.25%** of all riders already list their start-station ZIP as their postal code, indicating that it serves as a consistent and reliable geographic proxy. Applying this rule enhances spatial resolution and produces a standardized, station-based classification.

2.2.1.2 Sanity checks and temporal feature extraction

As part of data validation, we performed sanity checks to ensure internal consistency in the BlueBikes dataset. Trip duration was recomputed from standardized start and stop timestamps, and the results closely matched the original *Trip_Duration* feature, confirming accurate temporal recording **(Appendix, Table 2)**. To support detailed time-based analysis, timestamps were decomposed into start/stop dates and times, enabling examination of seasonal patterns, monthly trends, weekday effects, and time-of-day usage cycles key for identifying commuting peaks and broader shifts in rider behavior.

After running the full data-cleaning pipeline, correcting postal codes, extracting temporal features, and validating trip duration, the BlueBikes dataset was reduced to a curated set of variables ready for analysis **(Appendix Table 3)**. These refined variables form the foundation for all subsequent modeling and analysis.

2.2.1.3 Assigning city labels using USPS ZIP data

Because the BlueBikes dataset does not include city names for its ZIP codes, we incorporated a USPS ZIP–city reference table. After importing and cleaning the USPS file, i.e., standardizing fields and restricting entries to Massachusetts, we collapsed ZIP codes to obtain a unique ZIP–city lookup and merged it with the cleaned BlueBikes data **(Appendix Table 4)**. This assigned a validated physical city to each trip and enabled consistent geographic analysis across the six Greater Boston cities.

2.2.2 Rider type and temporal predictors

Rider type was coded as a two-level factor (Subscriber as the reference). To capture nonlinear temporal patterns, the start hour (0–23) was extracted and used as the continuous predictor in the degree-3 polynomial regression, enabling the model to reflect commuting peaks and leisure-time riding.

2.2.3 Final analytic sample

After all cleaning, the dataset was limited to Boston (main), Cambridge, Somerville, Brookline, and South Boston. These cities together corresponded to *2,531,441* observations. This curated sample is well-suited for assessing how rider type affects trip duration and how this relationship varies by time of day and location, forming the basis for the degree-3 polynomial regression model.

2.3 Statistical Model and model specification

To quantify whether rider type influences BlueBikes trip duration and how this relationship varies across time of day, we estimated a degree-3 polynomial regression model of the form.

$$Duration_i = \beta_0 + \beta_1 h_i + \beta_2 h_i^2 + \beta_3 h_i^3 + \beta_4 customer_i + \beta_5 (h_i \times customer_i) + \beta_6 (h_i^2 \times customer_i) + \beta_7 (h_i^3 \times customer_i) + \gamma_{c(i)} + \in_i$$

Where,
$Duration_i = trip\ duration\ for\ trip\ i$
$h_i = start\ hour\ of\ trip\ i$
$\text{Customer}_i = 1\ if\ rider\ is\ a\ customer, 0\ if\ subscriber$
$\gamma_{c(i)} = city\ fixed\ effect\ for\ the\ city\ in\ which\ trip\ i\ occurs$
$\varepsilon_i = error\ term$

Duration_i represents trip duration in minutes, h_i is the trip's start hour (0–23), and Customer_i indicates short-term riders, with Subscribers as the reference group. A cubic polynomial in start hour captures nonlinear variation in mean duration across the day, and interactions with rider type allow these temporal patterns to differ between Subscribers and Customers. City fixed effects γc(i) adjust for structural differences across Boston, Cambridge, Charlestown, Somerville, Brookline, and South Boston. The model was estimated to use ordinary least squares with classical standard errors and two-sided t-tests for inference.

3 Results

3.1 Regression result

(Table 2) details the coefficient estimates for the degree-3 polynomial regression. All coefficients are highly statistically significant ($p < 0.001$), which is expected given the large sample size ($N \approx 2.39$ million).

***Table 2**: Degree-3 Polynomial Regression of Trip Duration*

Coefficient	Estimate	Std.Error	t value	Pr (> |t|)
(Intercept)	18.4375	0.0392	469.99	<0.001
start_hour	–0.8618	0.0152	–56.88	<0.001
start_hour^2	0.1134	0.00151	75.03	<0.001
start_hour^3	–0.00332	0.0000386	–80.14	<0.001
rider_type1	–4.4141	0.0000414	–117.10	<0.001
cityBOSTON	0.1509	0.0377	8.76	<0.001
cityBROOKLINE	0.9044	0.0172	27.97	<0.001
cityCAMBRIDGE	–1.3504	0.0323	–79.45	<0.001
cityCHARLESTOWN	1.1917	0.0400	29.90	<0.001
citySOMERVILLE	–0.2324	0.0246	–9.43	<0.001
start_hour × rider_type1	0.3537	0.0152	23.35	<0.001
start_hour^2× rider_type1	–0.06395	0.00151	–42.32	<0.001
start_hour^3× rider_type1	0.00211	0.0000414	50.99	<0.001

Rider Type Effects. Customers take substantially longer trips than Subscribers. The coefficient of –4.41 minutes for rider_type1 (Subscribers), holding time and city constant, Customers ride about 4.4 minutes longer on average. This is consistent with leisure-oriented Customer behavior versus shorter, utilitarian Subscriber trips.

Temporal Pattern. The significant linear, quadratic, and cubic terms confirm a strong nonlinear daily pattern. Trip durations drop in the early morning, increase through midday, and flatten or rise slightly in the evening. The interaction coefficients show that Customers experience sharper midday increases and more pronounced curvature compared with Subscribers.

City Effects. Relative to the overall mean (sum-to-zero coding), Charlestown and Boston have longer average trip durations, while Cambridge and Somerville have shorter durations. These differences reflect geographic factors such as station density, street layout, and the dominant purpose of trips within each area.

3.2 Predicted duration profiles

(Figure 1.a) shows predicted trip duration over the 24-hour day for Subscribers and Customers across six regions. Curves are generated from the fitted degree-3 polynomial regression model and illustrate how temporal patterns in trip duration vary by rider type and city. Customers consistently take longer trips at every hour, with an 8–15-minute gap that reflects the strong rider-type effect in the model. Both groups display nonlinear temporal patterns, but the curve for Customers is much more pronounced, with sharp increases around midday and peaks in the late afternoon, especially Charlestown. Subscriber patterns are flatter and aligned with their expected commuting behavior (*i.e., only taking the necessary trips and relatively less leisure trips than the Customers*), showing shorter trips in the morning and modest increase later in the day. City differences mainly shift the overall level of the curves, but the gap between rider types

remains stable, reinforcing rider type as the dominant predictor of trip duration. Customer trips show monotonic increase in predicted mean duration during morning and midday intervals, followed by concave decrease in the evening, consistent with a non-linear time of day effect captured by the cubic specifications. Subscriber trips, however, display minimal variation across all intervals, indicating that the rider-type

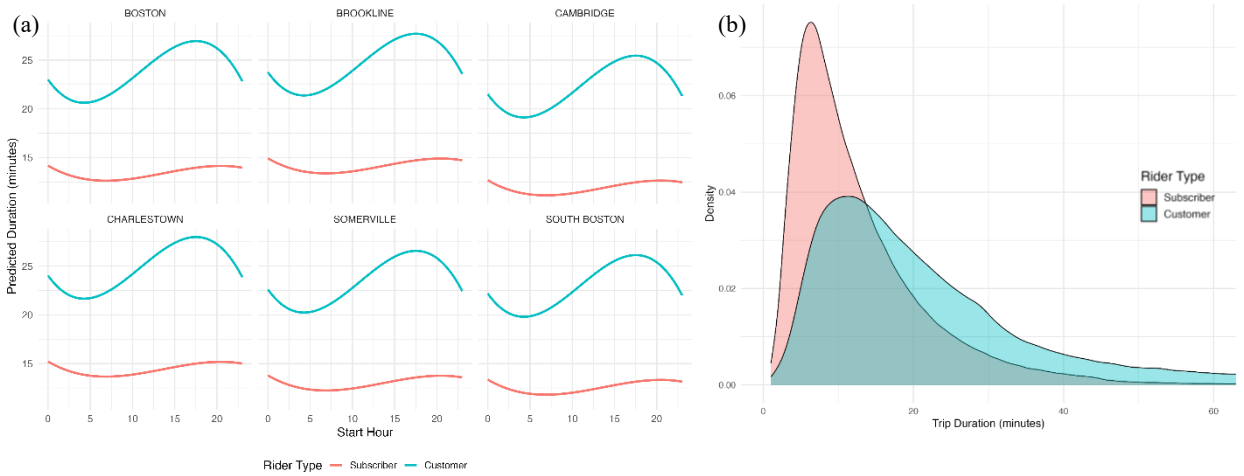


Figure 1: (a) predicted trip duration across the day by rider type and (b) Kernel density distribution of trip duration by rider type

interaction term absorbs most of the explainable variation.

4 Discussion and concluding remarks

Across the six regions, Cambridge and Somerville have the shortest average trip durations, while Charlestown shows the longest, reflecting differences in geography and station density **(Appendix, Table 5)**. Trip duration distributions further reveal clear rider-type contrasts: Subscriber rides cluster tightly around 5–15 minutes, whereas Customer trips display a long right tail extending past 45 minutes **(Figure 1.b)**. Empirical hourly averages indicate that Customer durations rise sharply from late morning into the afternoon peak, while Subscriber durations remain relatively flat with only slight morning dips patterns that motivate the model's cubic time-of-day specification **(Appendix, Figure 2)**. Finally, a heatmap of hourly trip counts shows strong daytime and evening ridership in Boston and Cambridge, whereas Charlestown has lower but steady usage throughout the day (**Appendix, Figure 3**).

The BlueBikes analysis significantly supports the literature research showing that nonlinear interactions across time, region, and rider attributes impact bikes haring behavior. Geographic disparities in duration reflect research suggesting that the built environment and urban structure influence bike ridership (Liu et al., 2025). Cambridge and Somerville's shorter, more consistent trip durations reflect their compact, high-density station networks, whereas Charlestown's longer and more variable trips align with nonlinear demand patterns driven by lower connectivity and more heterogeneous travel purposes.

This study shows that BlueBikes trip durations vary by rider type, time of day, and city, revealing clear policy implications. Longer customer trips and midday peaks point to policy needs such as flexible rebalancing and dynamic pricing to manage leisure demand. Shorter, commuter-focused morning trips underscore the policy importance of reliable bike availability and protected corridors. City-level differences in trip length also highlight where policy decisions on station expansion and connectivity improvements can most effectively advance equity and system performance.

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Appendix:

Table 1: Postal code classification (5 categories)

Postal source	Count	Percent (%)
Start station zip	986,413	26.25
End station zip	563,796	15.01
Valid but unmatched	1,198,138	31.89
Invalid and unmatched	543,297	14.46
Missing	465,637	12.39

Table 2: Comparison of original vs recalculated trip duration

Original trip duration	Recalculated trip duration
803	803.661
759	759.334
1314	1314.831
541	541.899
478	478.406
532	532.156

Table 3: Variables left after the exploratory data analysis on Bluebike dataset

tripduration	end.station.name	postal_source
starttime	end.station.latitude	postal.code_cleaned
stoptime	end.station.longitude	starttime_dt
start.station.id	bikeid	stoptime_dt
start.station.name	usertype	start_date
start.station.latitude	postal.code	stop_date
start.station.longitude	start_postal	start_time
end.station.id	end_postal	stop_time
trip_duration_calculated		

Table 4: Summary of missing USPS city information in the merged dataset

Metric	Value
Total rows	3,757,281
Missing physical city	470,252
Percent missing	12.52%

A. Top cities by trip volume

BlueBikes activity is heavily concentrated in a few cities: Cambridge and Boston together account for more than 1.9 million trips. A large portion of rides (470,252) fall under an unassigned city label, making “NA” the third-largest category and indicating gaps in ZIP–city metadata or geocoding. There are 1300 unique MA ZIP Codes in the entire BlueBikes dataset, and 854 unique ZIPs correspond to the observations labelled “NA.”

Beyond these groups, trip counts drop sharply, with Somerville, Brookline, and several smaller neighborhoods contributing modest but notable volumes. Overall, distribution shows strong geographic imbalance, with usage concentrated in core urban areas.

Table 5: Top 10 cities by total BlueBikes observation counts

City name	Observations
CAMBRIDGE	1,039,514
BOSTON	953,299
NA	470,252
SOMERVILLE	262,226
BROOKLINE	128,734
ROXBURY CROSSING	86,301
SOUTH BOSTON	82,654
CHARLESTOWN	78,735
JAMAICA PLAIN	71,599
BRIGHTON	71,559

Table 6: Distinct ZIP codes in the dataset and ZIPs associated with missing physical city.

Metric Count	Metric Count
Total Distinct ZIPs	1,334
Distinct ZIPs with Missing City	854

Table 7: Summary Statistics of Trip Duration by City

City	N	Mean Duration (min)	Median Duration (min)	SD Duration (min)
Boston	946,591	16.68	12.25	15.14
Brookline	128,046	16.69	13.27	13.37
Cambridge	1,035,705	14.00	10.19	12.55
Charlestown	78,209	17.83	14.13	14.50
Somerville	260,946	15.40	11.94	12.68
South Boston	81,944	16.99	13.08	14.22

Table 8: Descriptive statistics of trip duration by city and rider type

City	Rider type	Observations	Mean duration	Median duration	SD duration
Boston	Subscriber	688,345	13.29	10.52	10.32
Boston	Customer	258,246	25.72	19.27	21.07
Brookline	Subscriber	101,238	14.77	12.18	10.46
Brookline	Customer	26,808	23.97	18.42	19.34
Cambridge	Subscriber	865,109	12.12	9.23	9.56
Cambridge	Customer	170,596	23.57	17.76	19.58
Charlestown	Subscriber	56,433	14.92	12.62	10.40
Charlestown	Customer	21,776	25.37	20.04	19.90
Somerville	Subscriber	209,498	13.86	11.12	10.23
Somerville	Customer	51,448	21.67	16.21	18.45

South Boston	Subscriber	51,253	13.84	11.48	9.74
South Boston	Customer	30,691	22.26	16.27	18.37

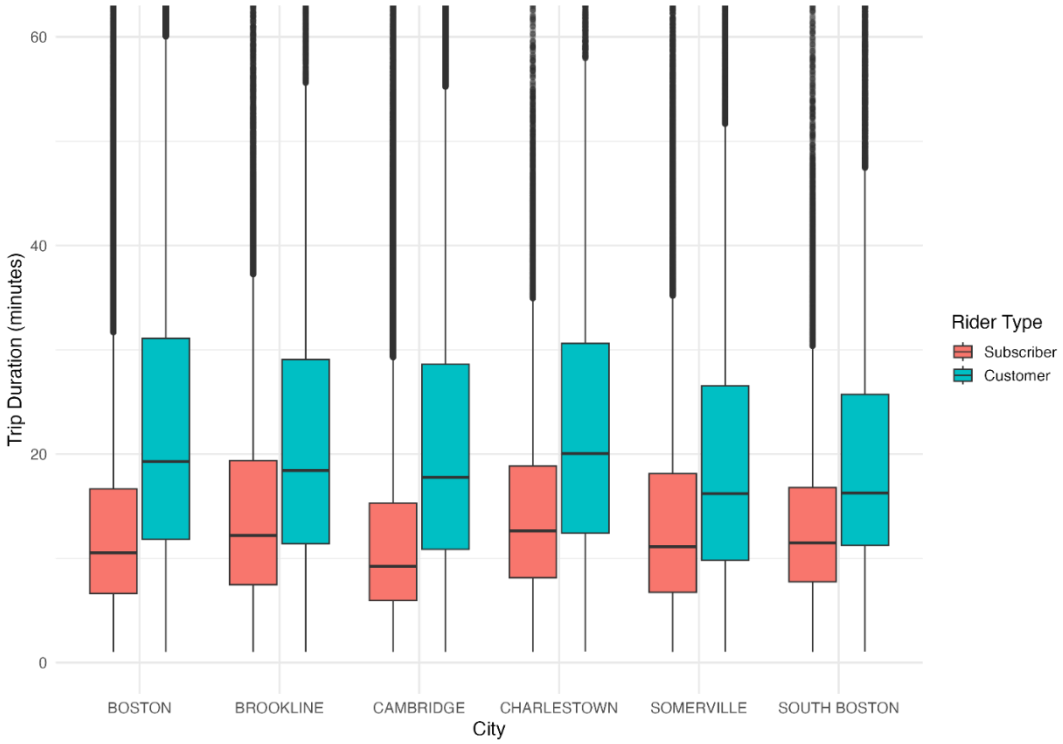


Figure 1: Trip duration distribution by city and rider type

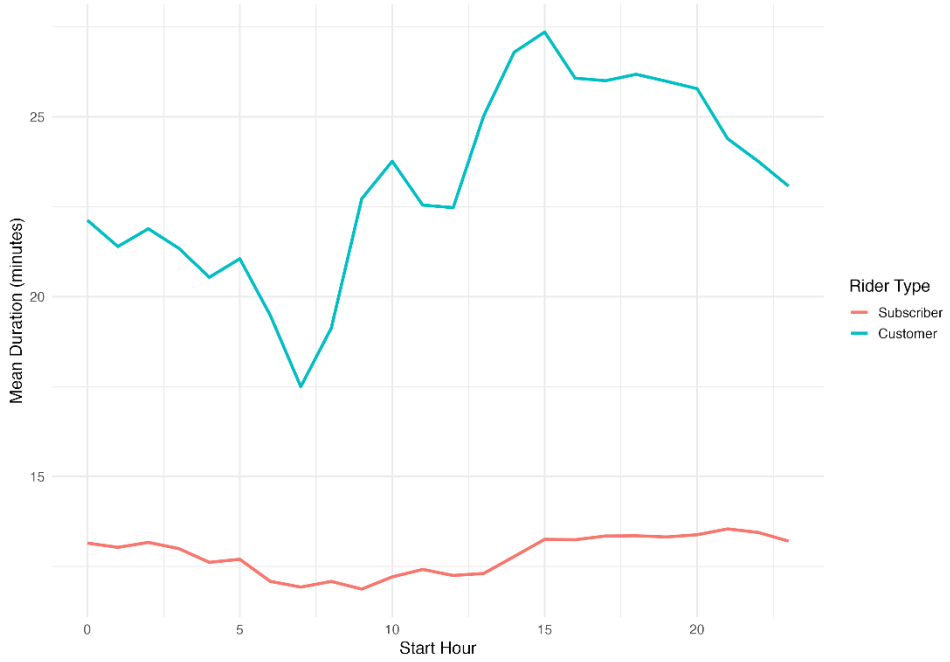


Figure 2: Raw hourly means trip duration by hour of day for Subscribers and customers

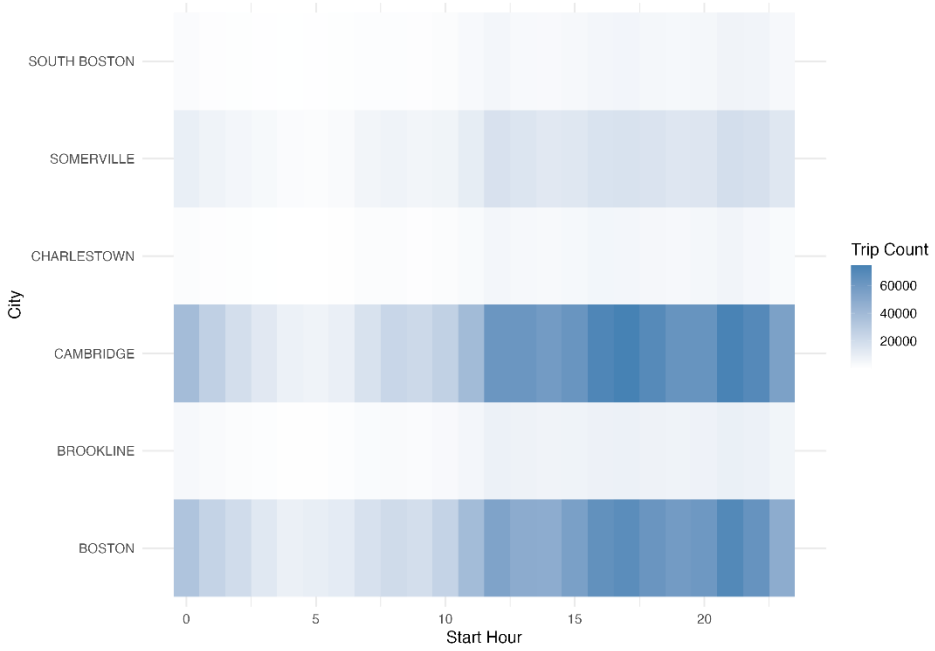


Figure 3:Heatmap of hourly trip volume across the six regions

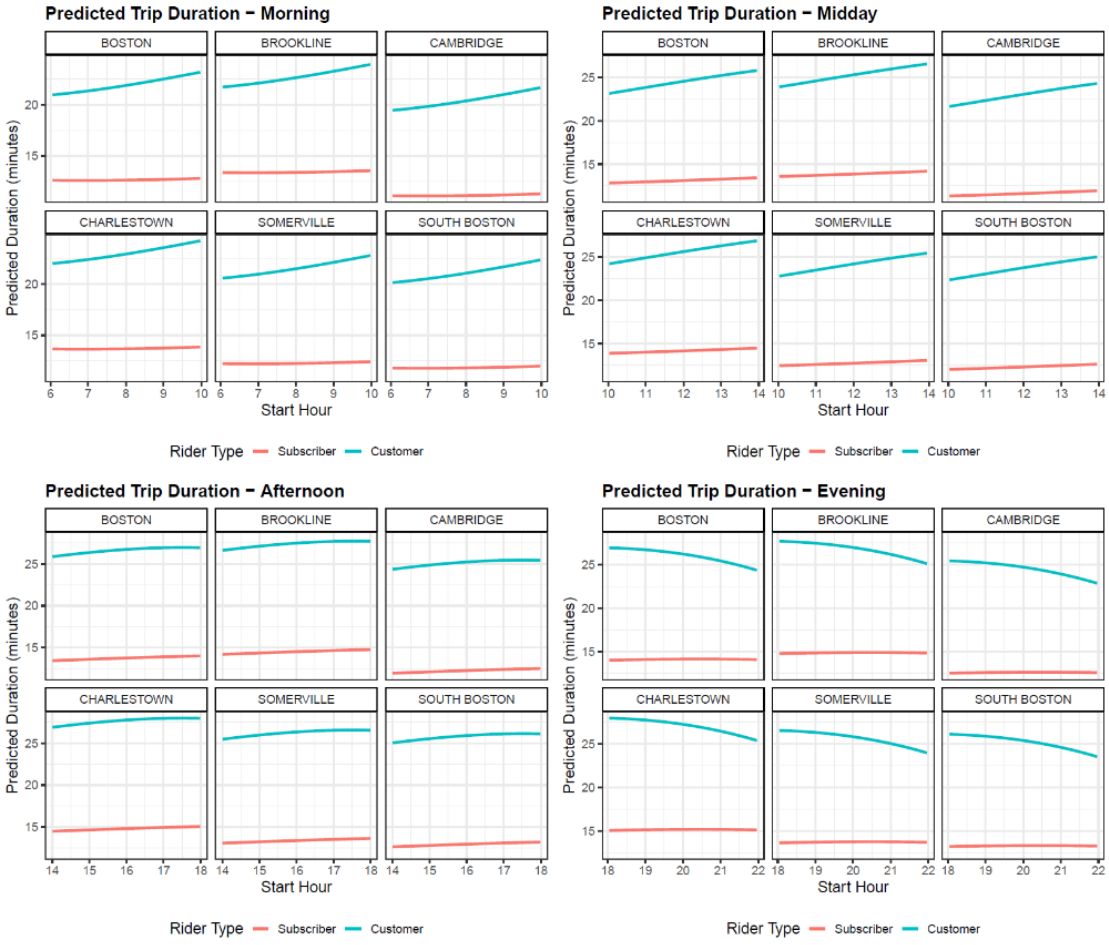


Figure 4: Predicted trip duration during morning, midday, afternoon and evening by rider type

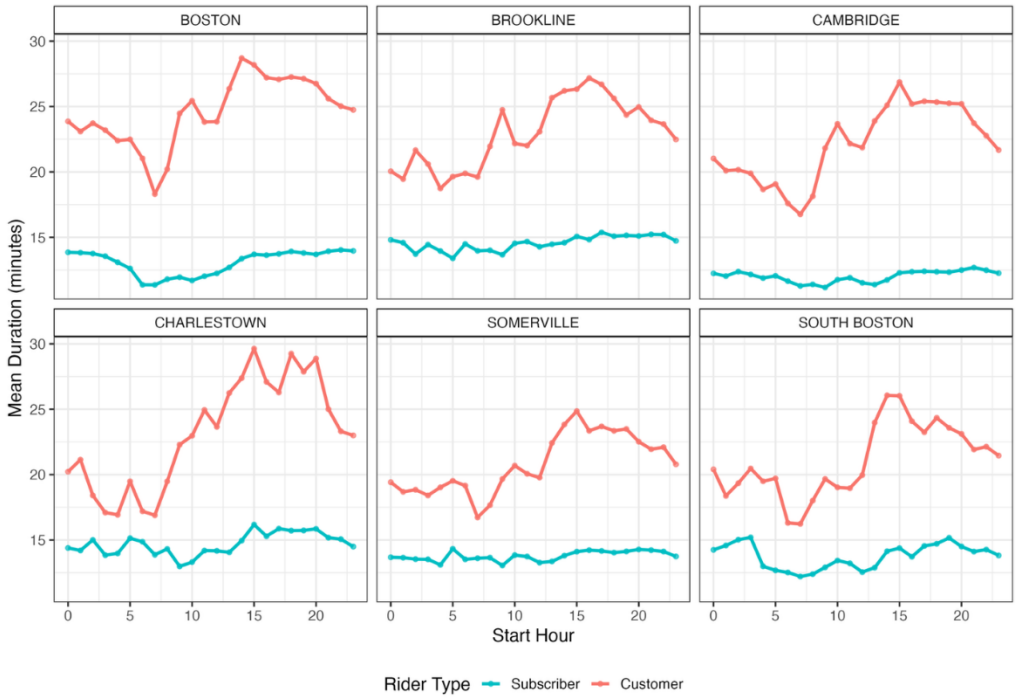


Figure 5: Raw trip duration across the day (fast smoothed curves)

B. Model diagnostics

Figures 6 and 7 provide diagnostic checks for the cubic regression model. The residuals vs. fitted plot shows increasing variance at higher fitted values, a characteristic pattern in transportation data where leisure trips produce more variability than short commuter trips. The Q–Q plot reveals right-tail deviations from normality, consistent with the heavy-tailed distribution of Customer ride durations. Although these patterns indicate imperfect adherence to normality assumptions, the model remains appropriate for mean-structure inference given the large sample size.

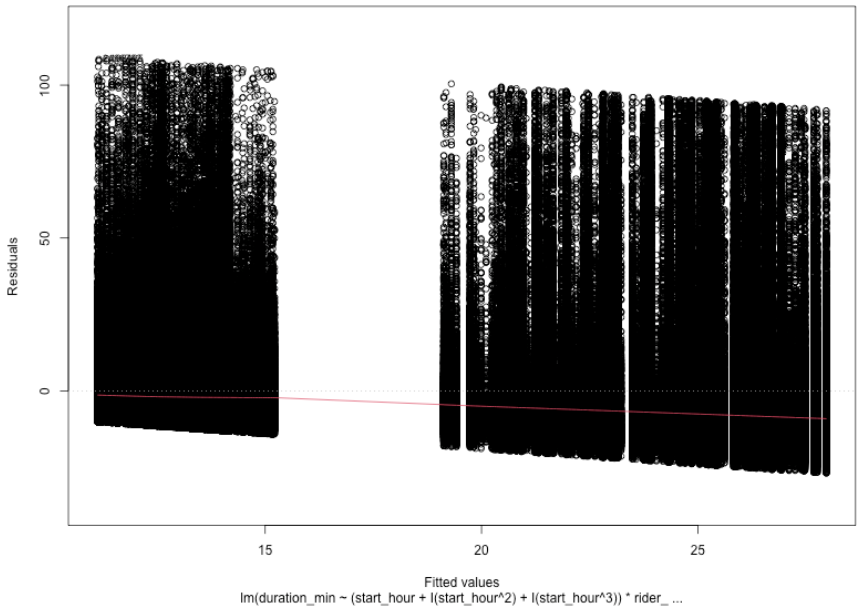


Figure 6: Residuals vs. fitted values for the cubic regression model

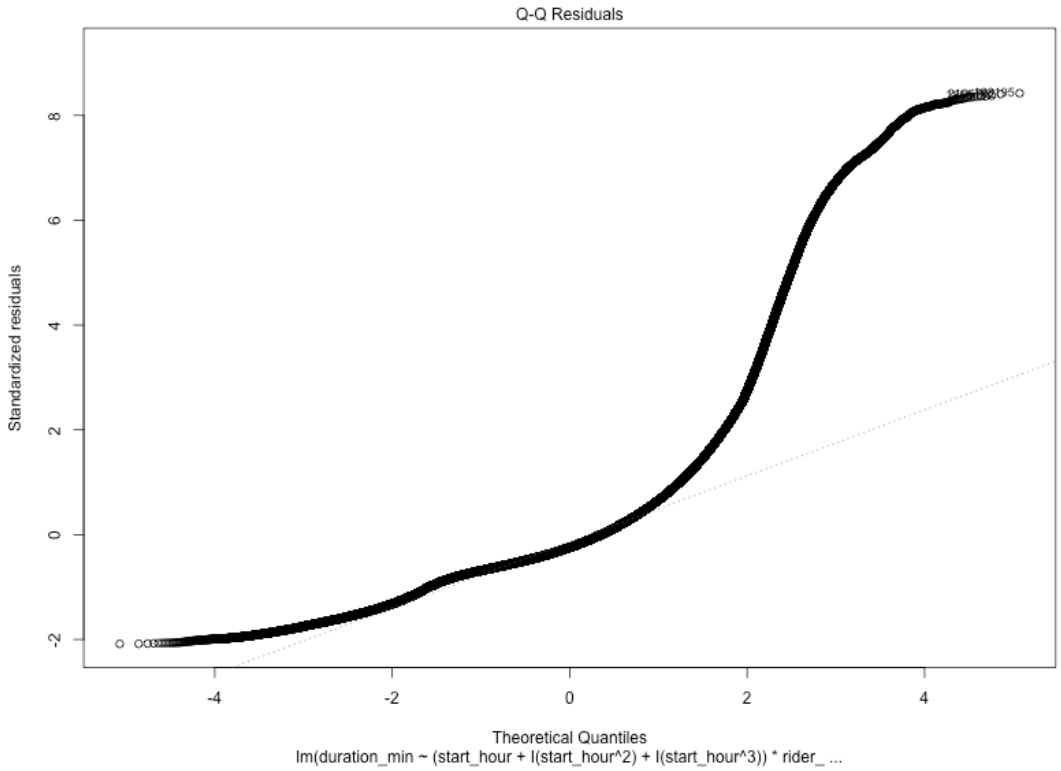


Figure 7: Q-Q plot of standardized residuals from the cubic regression model

C. R programming codes

```
############################################################
# STEP 1 — LOAD CLEANED BLUEBIKES DATA
############################################################

bluebikes_final <- read.csv(
  data_path,
  header = TRUE,
  stringsAsFactors = FALSE
)

cat("Rows in bluebikes_final:", nrow(bluebikes_final), "\n")

############################################################
# STEP 2 — BUILD MODELING DATASET (analysis_df)
############################################################

analysis_df <- bluebikes_final %>%
  mutate(
    # Parse timestamps
    start_dt = parse_date_time(starttime, orders = c("ymd HMS", "Ymd HMS"),
                               tz = "America/New_York"),
    stop_dt  = parse_date_time(stoptime,  orders = c("ymd HMS", "Ymd HMS"),
                               tz = "America/New_York"),
    # Duration in minutes
    duration_min = as.numeric(stop_dt - start_dt, units = "secs") / 60,
    # Hour of day
    start_hour = hour(start_dt),
    # Rider type factor
    rider_type = factor(usertype, levels = c("Subscriber", "Customer")),
    # City factor using cleaned city name
    city = factor(PHYSICAL_CITY_cleaned)
  ) %>%
  # Analysis restrictions
  filter(
    !is.na(start_dt),
    !is.na(stop_dt),
    !is.na(duration_min),
    duration_min > 0,
    duration_min <= 120,                              # cap at 2 hours
    !is.na(rider_type),
    !is.na(city),
    city %in% c("BROOKLINE", "CAMBRIDGE", "SOMERVILLE",
                "BOSTON", "SOUTH BOSTON", "CHARLESTOWN")
  )
```

```r
cat("Rows in analysis_df:", nrow(analysis_df), "\n")
str(analysis_df)

############################################################
# STEP 3 — SUMMARY TABLE (CITY × RIDER TYPE)
############################################################

summary_table <- analysis_df %>%
  group_by(city, rider_type) %>%
  summarise(
    n = n(),
    mean_duration   = round(mean(duration_min), 2),
    median_duration = round(median(duration_min), 2),
    sd_duration     = round(sd(duration_min), 2),
    .groups = "drop"
  )

print(summary_table)

# (If you ever want LaTeX for this table:)
kable(
  summary_table,
  caption = "Descriptive Statistics of Trip Duration by City and Rider Type",
  align   = "lccc"
)

############################################################
# STEP 4 — FIT DEGREE-3 POLYNOMIAL REGRESSION MODEL
############################################################

model_poly3 <- lm(
  duration_min ~
    (start_hour + I(start_hour^2) + I(start_hour^3)) * rider_type +
    city,
  data = analysis_df
)

summary(model_poly3)

############################################################
# STEP 5 — PREDICTION GRID + POLY CURVES PLOT
############################################################

# Extract factor levels from the model (guarantees compatibility)
valid_cities  <- levels(model_poly3$model$city)
valid_riders  <- levels(model_poly3$model$rider_type)

# Build prediction grid for 0–23 hours, both rider types, all cities
hour_grid <- expand.grid(
  start_hour = seq(0, 23, length.out = 200),
  rider_type = valid_riders,
  city       = valid_cities
)

# Predicted duration
hour_grid$pred <- predict(model_poly3, newdata = hour_grid)

# Plot model-predicted curves
poly_plot <- ggplot(hour_grid,
                    aes(x = start_hour, y = pred, color = rider_type)) +
  geom_line(size = 1.2) +
  facet_wrap(~ city) +
  labs(
    title  = "Predicted Trip Duration Across the Day by Rider Type and City",
    x      = "Start Hour",
    y      = "Predicted Duration (minutes)",
    color  = "Rider Type"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    plot.title      = element_text(face = "bold", size = 16)
  )

ggsave(
  filename = paste0(outdir, "poly_curves.png"),
  plot     = poly_plot,
  width    = 11,
  height   = 8,
  dpi      = 300
)
```

```r
############################################################
# STEP 6 — ADDITIONAL VISUALS (EXHIBITS)
############################################################

# 6.1 Duration distribution by rider type
p_density <- ggplot(analysis_df,
                    aes(x = duration_min, fill = rider_type)) +
  geom_density(alpha = 0.4) +
  coord_cartesian(xlim = c(0, 60)) +
  labs(
    title = "Distribution of Trip Duration by Rider Type",
    x     = "Trip Duration (minutes)",
    y     = "Density",
    fill  = "Rider Type"
  ) +
  theme_minimal(base_size = 14)

ggsave(
  filename = paste0(outdir, "Exhibitsduration_density.png"),
  plot     = p_density,
  width    = 11,
  height   = 8,
  dpi      = 300
)

# 6.2 Empirical mean duration by hour (raw averages)
raw_hourly <- analysis_df %>%
  group_by(start_hour, rider_type) %>%
  summarise(mean_duration = mean(duration_min), .groups = "drop")

p_raw_means <- ggplot(raw_hourly,
                      aes(x = start_hour, y = mean_duration,
                          color = rider_type)) +
  geom_line(size = 1.2) +
  labs(
    title = "Empirical Mean Trip Duration by Hour of Day",
    x     = "Start Hour",
    y     = "Mean Duration (minutes)",
    color = "Rider Type"
  ) +
  theme_minimal(base_size = 14)
```

```r
ggsave(
  filename = paste0(outdir, "Exhibitsraw_hourly_means.png"),
  plot     = p_raw_means,
  width    = 11,
  height   = 8,
  dpi      = 300
)

# 6.3 Boxplots by city and rider type
p_box <- ggplot(analysis_df,
                aes(x = city, y = duration_min, fill = rider_type)) +
  geom_boxplot(outlier.alpha = 0.2) +
  coord_cartesian(ylim = c(0, 60)) +
  labs(
    title = "Trip Duration Distribution by City and Rider Type",
    x     = "City",
    y     = "Trip Duration (minutes)",
    fill  = "Rider Type"
  ) +
  theme_minimal(base_size = 14)

ggsave(
  filename = paste0(outdir, "Exhibitscity_boxplots.png"),
  plot     = p_box,
  width    = 11,
  height   = 8,
  dpi      = 300
)

# 6.4 Overall city summary table -> saved as PNG
city_table <- analysis_df %>%
  group_by(city) %>%
  summarise(
    n               = n(),
    mean_duration   = round(mean(duration_min), 2),
    median_duration= round(median(duration_min), 2),
    sd_duration     = round(sd(duration_min), 2),
    .groups         = "drop"
  )

png(filename = paste0(outdir, "Exhibitscity_summary_table.png"),
    width = 1000, height = 400)
grid.table(city_table)
dev.off()

# 6.5 Hour × city heatmap of trip counts
heat_df <- analysis_df %>%
  group_by(city, start_hour) %>%
  summarise(count = n(), .groups = "drop")
```

```r
p_heat <- ggplot(heat_df,
                 aes(x = start_hour, y = city, fill = count)) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "steelblue") +
  labs(
    title = "Trip Count by City and Hour of Day",
    x     = "Start Hour",
    y     = "City",
    fill  = "Trip Count"
  ) +
  theme_minimal(base_size = 14)

ggsave(
  filename = paste0(outdir, "Exhibitshour_city_heatmap.png"),
  plot     = p_heat,
  width    = 11,
  height   = 8,
  dpi      = 300
)

############################################################
# STEP 7 – MODEL DIAGNOSTICS (RESIDUALS + QQ PLOT)
############################################################

# Residuals vs fitted
png(filename = paste0(outdir, "Exhibitsmodel_residuals_vs_fitted.png"),
    width = 800, height = 600)
plot(model_poly3, which = 1)  # residuals vs fitted
dev.off()

# Q-Q plot
png(filename = paste0(outdir, "Exhibitsmodel_qqplot.png"),
    width = 800, height = 600)
plot(model_poly3, which = 2)  # QQ-plot
dev.off()

cat("All analysis and Exhibits generated.\n")
```